

Simulated free-flight validation — run checklist

Only the identifiable cases fly; S10 and S12 tip under the excitation collective and are excluded by construction. Per trial: set the case offset and mass, configure the feedforward from the *identified* offset (w_ff only), arm, take off to the standard hover point, hold ≥ 5 s, stop the log. Same topics as the excitation runs; bag name <variant>_<trial> under S<k>/.

case	truth [mm]	identified [mm]	$\bar{M}_{ff}$ (M_x, M_y) [N·m]	m [kg]	wo_ff $\times 3$	w_ff $\times 3$
S1	(−6, +0)	(−6.01, −0.11)	(−0.003, +0.181)	3.066	☐ ☐ ☐	☐ ☐ ☐
S2	(+0, +10)	(−0.01, +10.00)	(+0.301, +0.000)	3.066	☐ ☐ ☐	☐ ☐ ☐
S3	(+10, −5)	(+10.02, −5.05)	(−0.152, −0.301)	3.066	☐ ☐ ☐	☐ ☐ ☐
S4	(+20, +20)	(+20.19, +20.19)	(+0.607, −0.607)	3.066	☐ ☐ ☐	☐ ☐ ☐
S5	(+20, −20)	(+20.22, −20.16)	(−0.606, −0.608)	3.066	☐ ☐ ☐	☐ ☐ ☐
S6	(−20, +20)	(−20.26, +20.09)	(+0.604, +0.609)	3.066	☐ ☐ ☐	☐ ☐ ☐
S7	(−20, −20)	(−20.21, −20.24)	(−0.609, +0.608)	3.066	☐ ☐ ☐	☐ ☐ ☐
S8	(+25, +25)	(+25.22, +25.20)	(+0.758, −0.759)	3.066	☐ ☐ ☐	☐ ☐ ☐
S9	(+32, +32)	(+32.37, +32.09)	(+0.965, −0.974)	3.066	☐ ☐ ☐	☐ ☐ ☐
S11	(+38, +14)	(+38.13, +14.22)	(+0.428, −1.147)	3.066	☐ ☐ ☐	☐ ☐ ☐
S13	(+25, +25)	(+25.35, +25.23)	(+0.797, −0.801)	3.220	☐ ☐ ☐	☐ ☐ ☐

66 flights = 11 cases $\times$ 2 variants $\times$ 3 trials. The identified offsets are the R3 excitation-campaign estimates, frozen in `analysis/freeflight_sim_checklist.py`. $\bar{M}_{ff}$ is the pivot-free direction-pair average of that campaign, $\bar{M}_{ff} = \frac{1}{2}(\bar{M}_{\text{crit},+} + \bar{M}_{\text{crit},-}) = (W\hat{y}, -W\hat{x})$ – the moment the w_ff feedforward must cancel.